

Chapter-7

7. Demographic Methods and Health Services Statistics

7.1 Definition

Demography: is composed of two Greek words, namely, ' **Demos**' meaning **people** and **Graphien** meaning **to describe**. Therefore, demography means description of people or the scientific study of human population.

The focuses of demography are

- **Size:** is the number of persons in a population at a given time.
- **composition (structure):** is the distribution of a population into its various groupings mainly by age and sex.
- **Distribution:** is the arrangement of the population in the territory of a nation in geographical, residential area, climatic zone, etc.
- **Change:** refers to the increase or decline of the total population or its components. The components of change are: Birth(fertility) ,Death(Mortality) and Migration(In and Out)

7.2 Sources of Demographic data: Demographic information can be acquired through two main ways:

- a. Enumeration Method : Census and Survey
- b. Registration Method: Vital registration system (VRS)

Census: There are two main different schemes for enumerating peoples in a census.

Dejure: enumeration or count is done according to the usual or legal residence (permanent residence). A dejure count of the members of a household excludes temporary residents and visitors, but includes permanent residents who are temporarily away.

Advantage

Dejure: yields information relatively unaffected by the seasonal fluctuations.

Disadvantages

- Some persons may be omitted from the count while some others may be counted twice.
- In some situations, it is difficult to be sure just which is a person's usual or legal residence.
- Information collected regarding persons away from home is often incomplete or incorrect.

Defacto: enumeration is done according to the **actual place of residence** on the day of the census.

A defacto count includes temporary residents and visitors, but excludes permanent residents who happen to be away on the day of the census.

Advantage

Defacto: offers less chance of double counting and omission.

Disadvantages

- Population figures may be inflated or deflated by tourists, traveling salesmen, and other transients.
- In areas with high migration, the registration of vital events (vital statistics) is subject (liable) to distortion.

Characteristics of a Census

- Individual enumeration.
- Universality within defined territory (It should refer to the people inhabiting a well-defined territory).
- Simultaneity (enumeration with respect to a well-defined point in time).
- Defined periodicity (It should be taken at regular intervals usually ten years).

Census operation

The entire census operation has 3 stages:

- Pre- enumeration planning and preparatory work.
- Enumeration field work (data collection)
- Post- enumeration editing, coding, compilation, tabulation, analysis, and publication of results.

Uses of a census

- gives complete and valid picture of the population composition and characteristics
- Serves as a sampling frame.
- Provides with vital statistics of the population in terms of fertility and mortality.
- Census data are utilized in a number of ways for planning the welfare of the people.

Survey: Surveys are made at a given moment, in specific territory without periodicity to obtain information from part of the population which is considered to be representative.

Vital registration system:

The recording of vital events (births, deaths, marriage, divorce, adoption, etc) on continuous bases.

Note that VRS is a **continuous process** while **census** is a **one time activity**.

Characteristics of a VRS

- Comprehensiveness (All or every vital events).
- Compulsory by law.
- Compilation done centrally.
- Registration is a continuous process.

7.2.1 Population composition

Age-sex composition

Age and sex are the most important/ basic characteristics of a population. Age –sex composition of a population refers to the number or proportion of males and females on each age group.

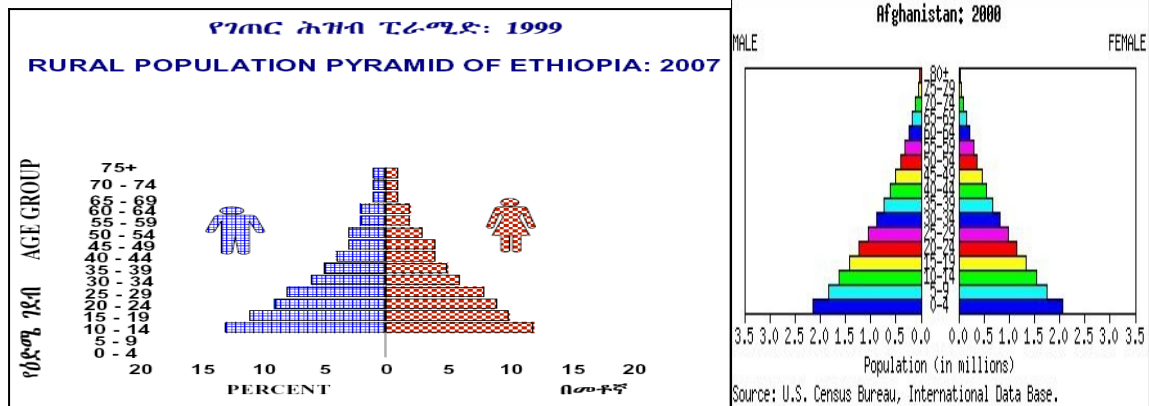
Every population has a different age and sex composition.

Population pyramid

Population pyramid is a graphical display of age and sex composition at a given time. It shows numbers or proportions of males and females in each age group. Conventionally females are shown on the right of the pyramid and young persons are displayed at the bottom of the pyramid. The pyramid can be constructed by first constructing a histogram separately for both sexes and superimposing these histograms vertically for the corresponding age group with the histogram constructed for the female population being on the right side. The shape of the pyramid is the direct or indirect effect of the influence of the past events such as births, deaths or migration.

Expansive or expanding

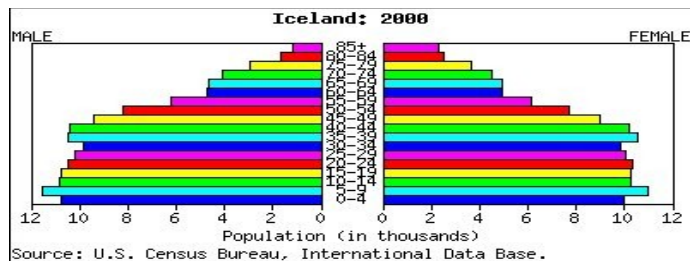
- ☐ Characterized by wide base indicating a high-birth rate & the narrow top indicates a high death rate.
- ☐ It is a characteristic of a lower standard of living:
 - High birth rate due to poor access to birth control, lack of education etc.;
 - High death rate due to poor medical care & nutrition.



Stationary or Stable Population

- ☐ It has a half ellipse shape.
- ☐ The base of the pyramid is similar in width to the population of the reproductive ages which indicates a stable population.
- ☐ It is the characteristic of a high standard of living due to:
 - ☒ Low birth rate due to good family planning, access to birth control, financial planning, education, etc.;

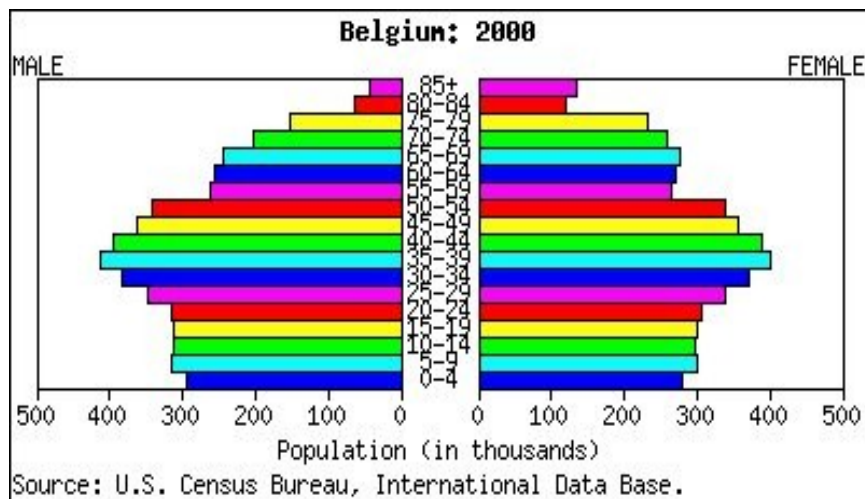
Low death rate due to good medical care, nutrition, education etc



. Contractive or contracting

- ☐ Contractive or contracting population pyramids have a narrower base than the reproductive age population.
- ☐ This indicates a decreasing population trend.

The low birth rate is indicative of a well developed country



7.3 Stages in Demographic Transition

Demographic transition refers to the change that populations undergo from high rates of births and deaths to low rates of births and deaths. The theory of demographic transition attempts to explain the changes in mortality and fertility in three different stages therefore we distinguish among three types of pyramid.

a. pre-transitional: characterized by high mortality and high fertility, with low (moderate) population growth (young population). It can be shown by triangular, broad based pattern reflecting the high birth rates. This type of population is found in primitive societies and is sometimes known as expansive type I/stage I. (expansive pyramid).

b. Transitional: Characterized by high birth rate and reduced death rates, with high (rapid) growth rate (young population). The drop in the death rate is usually brought about by improved medical care following socio economic development. The shape of the population pyramid is triangular characterizing a developing society. Sometimes known as expansive type II/stage II.

c. post-transitional: characterized by low birth and death rates with stable, moderate growth rate. Narrow based pyramid and steeper sides. This is typical of advanced or developed countries and is sometimes known as stationary type III/stage III. (stationary pyramid)

NOTE: When constructing a pyramid using a percentage, take the percentage from the group total.

Vital Statistics

Rate:

- ✚ Usually reserved to refer to those calculations that imply the **probability** of the occurrence of some event.
- ✚ A **proportion** with a time element. A rate is expressed in the form

$$\left(\frac{a}{a+b} \right) * k \quad \text{Where}$$

a= the frequency with which an event has occurred during some specified period of time.

a+b= the **maximum number** of persons exposed to the risk of the event during the same period of time. Sometimes referred to as the **population at risk**.

K=some number a multiple of 10(10, 100 ...).used to facilitate the comprehension of rates.

Note that the numerator is included in the denominator.

Rates could be **Crude** or **Specific**.

Crude⇒ - calculated for the **entire population**.

- affected by the structure of the population.

Specific ⇒ calculated for a **particular category**.

Ratio: is a fraction of the form $\left(\frac{c}{d} \right) * k$ Where c and d are the frequency of occurrence of some event or item.

NB: The numerator need not be a part of the denominator.

- ✚ All rates are ratios.
- ✚ All ratios are not rates.
- ✚ Ratio does not relate to a particular time.
- ✚ No restrictions on range or dimensions.

Percentage and Probability

Percentage is a special type of proportion in which the ratio is expressed per 100 and **probability** is similar to rate but the denominator is the population at the beginning of the period of observation.

1. **Sex ratio (SR):** is the total number of male population per 100 female populations.

$$SR = \frac{M}{F} * 100$$

Where M = Male population and F = Female population.

SR is used for purpose of comparison

- ✚ The balance between the two sexes.
- ✚ The variation in the sex balance at different ages.

2. **Child-woman ratio (CWR):** is the ratio of the number of children under 5 years of age to the number of women in the reproductive age usually 15 to 49.

$$CWR = \left(\frac{P_{0-4}}{p_{15-49}^f} \right) * 1000$$

= Number of children under 5 years of age per 1000 women in the child bearing age.

3. **Dependency ratio (DR):** It describes the relation between the potentially self-supporting portion of the population and the dependent portions at the extremes of age.

$$DR = \left(\frac{P_{0-14} + P_{65+}}{P_{15-64}} \right) \times 100$$

$$DR = \left(\frac{p_{0-14}}{p_{15-64}} * 100 \right) + \left(\frac{p_{65+}}{P_{15-64}} * 100 \right)$$

⇓

⇓

$$= CDR + ADR$$

Where CDR is child dependency ratio and ADR is aged dependency ratio.

7.4 Measure of Fertility and Measure of Mortality

7.4.1 Measures of fertility

Fertility: refers to the actual reproductive performance.

1. **Crude birth rate (CBR):** is the number of **live births** in a year per 1000 mid year population in the same year.

$$CBR = \frac{\text{Total number of live birth in a year}}{\text{Mid-year population in the same year}} \times 1000$$

Live Birth: is the complete expulsion or extraction of a fetus from its mother as a product of conception irrespective of the duration of pregnancy, which after such separation show evidence of life (like breathing, pulsation of heart, etc).

2. **General fertility rate (GFR)**: is the number of births in a specified period per 1000 women in aged 15 to 49.

$$GFR = \frac{\text{Total number of live births during a year}}{\text{Mid-year female population in the child bearing age}} \times 1000$$

Compared to CBR, GFR approximates the number of persons actually exposed to the risk of bearing a child.

The rate of childbearing is not the same or uniform throughout the child bearing ages, a rate that permits the analysis of fertility rates for shorter maternal ages is desirable. The rate used is called Age specific fertility rate (ASFR).

3. **Age Specific Fertility Rate (ASFR)**: is the number of live births in a specified period per 1000 women of a given age or age group.

$$ASFR = \frac{\text{Number of births to women of a certain age in a year}}{\text{Total number of women in the specified age}} \times 1000$$

4. **Total Fertility Rate (TFR)**

- ✚ If the age-specific rates for all ages are added and multiplied by the interval into which the ages were grouped, the result is called the total fertility rate.
- ✚ The resulting figure is an estimate of the number of children a cohort of 1000 women would have if, during their reproductive years, they reproduced at the rates represented by the age-specific fertility rates from which the total fertility rate is computed.

$$TFR = w \sum_{i=1}^k \frac{B_i}{P_i} * 1000$$

The assumption in the calculation is that the women in the reproductive age are classified in to 'k' classes of width 'w'.

5. **Cumulative Fertility Rate (CFR)**: The CFR is calculated /computed in the same manner as the TFR except that the adding process can terminate at the end of any desired group. CFR measures the total number of children a cohort of women will have at the end of any desired reproductive age.

6. **Gross Reproduction Rate (GRR):** Is the total fertility rate (TFR) restricted to female births only.

$$\text{GRR} = \sum_{i=1}^k \frac{B_i^f}{P_i^f} * 1000$$

But if we do not have female births and female population by age we compute GRR by

$$\text{GRR} = \frac{B^f}{B^t} * \text{TFR}$$

Where $B^t = B^m + B^f$

$$\Rightarrow \frac{B^f}{B^t} = \frac{B^f}{B^f + B^m} = \frac{1}{1 + \frac{B^m}{B^f}} = \frac{1}{1 + \text{SRb}}$$

where SRb is sex ratio at birth.

$$\text{Therefore, GRR} = \frac{1}{1 + \text{SRb}} \text{TFR}$$

GRR gives the average number of daughters a cohort of women would have at the end of reproduction, in the absence of mortality.

7. **Net Reproduction Rate (NRR)** measures the extent to which the females in the reproductive age-groups are replacing themselves in the next generation. The net reproduction rate is 1 in the stationery population. NRR is less than one in declining population and greater than one in growing population.

7.4.2 Measures of Mortality

Mortality: refers to death or permanent disappearance (deduction) of a member from the population.

1. **CRUDE DEATH RATE(CDR)**
CDR is defined as

$$\text{CDR} = \frac{\text{Total number of deaths during a year}}{\text{Total population}} * 1000$$

CDR measures the rate at which deaths are taking place in a given population during a specified year.

2. **Age specific death rates(ASDR)**

ASDR is defined as the total number of deaths occurring in a specified age group of the population per 1000 mid year population of the same age group.

$$ASDR_a = \frac{\text{Total deaths at age or group a}}{\text{Mid – year population at age or group a}} \times 1000$$

3. **CAUSE SPECIFIC DEATH RATIO OR RATE (CSD)**

A cause specific death ratio (Proportionate mortality ratio) represents the percent of all deaths due to a particular cause or group of causes.

$$\text{CSD ratio for cause } i = \frac{D_i}{D_t} * 1000$$

Where D_i is the total death from cause i and a D_t is the total deaths from all causes.

Cause specific death rate (CSDR) is the number of deaths from cause i during a year per 1000 of mid-year population.

$$CSDR = \frac{\text{Total deaths from a given cause}}{\text{Population at risk}} \times 1000$$

4. **Infant mortality rate (IMR)**

IMR measures the risk of dying during infancy (the first age of life), and is defined as

$$IMR = \frac{\text{Deaths of children under 1 year of age}}{\text{Total live birth}} \times 1000$$

IMR is the ‘**probability**’ of dying between birth and age one year per 1000 live births.

Neonatal Mortality Rate (NMR): Measures the risk of dying within 28 days of birth.

Post Neonatal Mortality Rate (PNMR): Measures the risk of dying during infancy after the first 4 weeks of life.

5. **Child Death rate (ChDR):** measures the mortality condition among children between 1 to 4 years of age.

$$ChDR = \frac{\text{Deaths of children in 1-4 year age group}}{\text{Mid-year population of children in 1-4 year age group}} \times 1000$$

6. **Maternal Mortality Rate (MMR)**

MMR is defined as the number of deaths of mothers (DM) due to maternal causes, that is, complications of pregnancy, child birth, etc per 1000 live births during a year.

$$\text{MMR} = \frac{\text{deaths of mothers due to maternal causes in a year}}{\text{Total live births in the same year}} \times 1000$$

7. **UNDER 5 MORTALITY RATE**: - the probability of dying between birth and age 5 per 1000 live births

$$\text{5 Mortality} = \frac{\text{Total deaths of under 5 children}}{\text{total live birth}}$$

8. **CHILD MORTALITY RATE**: - the probability of dying between age 1 and 5 year per 1000 children on age one year.

$$\text{CMR} = \frac{\text{Deaths of children between age 1 and 5 years}}{\text{Total number of children age 1 year}} \times 1000$$

7.5 Population Change and Doubling time

Population change has three components

- *Birth*
- *Death*
- *Migration (In - Out)*

Population change = (Birth-Death) + (In migration-Out migration)

Rate of population growth(r) = CBR-CDR. This rate is sometimes called **crude rate of natural increase** because its computation is based on crude rates.

7.5.1 Population Projection:-is a statement about what the size and age composition of the population would be at a given future time if the fertility, mortality and migration were to follow certain specific trends. Based on this rate of increase(r), the population (P_t) of an area with current population size of (P_0) can be projected at some time t in the short time interval (mostly not more than 5 years) using the following formula.

$$P_t = P_0 (1 + r)^t \text{ OR } P_t = P_0 e^{rt} \text{ — the exponential projection formula}$$

7.5.2 Population doubling time

The doubling time of the size of population can be estimated based on the formula for projecting the population.

From the formula we want to find out the time t at which the current population P_0 will be $2 P_0$. Therefore,

$$P_t = P_o (1 + r)^t \text{ implies } 2 P_o = P_o (1 + r)^t \Rightarrow 2 = (1 + r)^t \Rightarrow \ln 2 = t \ln(1 + r)$$

$$\text{Hence, } t = \frac{\ln 2}{\ln(1 + r)}$$

A more practical approach to calculate the population doubling time is:

$$2 P_o = P_o e^{rt} \Rightarrow 2 = e^{rt} \Rightarrow \ln 2 = rt \Rightarrow t = \frac{\ln 2}{r} = \frac{0.7}{r}$$

Example

The following data were obtained from the 1984 population and housing census of Illubabor region.

Total population = 975,658

Total male population = 484,637

Total female population = 491,021

< 1 year population = 3.3% of the total population

Total number of births occurred during the last 12 months before the census day = 36,813

Total number of deaths occurred during the last 12 months before the census day = 10,462

Of the deaths 2,303 were occurred to those who are below one year of age.

No of deaths of mothers due to maternal conditions 12 months before the census day = 300.

Sex ratio at birth = 103.2

Table 1: Percentage distribution of enumerated population of Illubabor region by age and sex, 1984 population census

Age group	Males	Females	Total
0 – 4	18.6	18.2	18.4
5 - 9	18.7	17.6	18.2
10 - 14	10.7	10.1	10.4

15 - 19	8.4	6.8	7.6
20 - 24	6.5	5.7	6.5
25 - 29	6.1	6.9	6.5
30 - 34	5.8	6.2	6.0
35 - 39	5.0	5.9	5.4
40 - 44	4.2	4.7	4.4
45 - 49	3.7	3.5	3.6
50 - 54	3.0	3.8	3.4
55 - 59	2.3	2.3	2.3
60 - 64	3.0	3.9	3.5
65+	5.0	4.4	4.5
Total	100.0	100.0	100.0

Table 2. Distribution of women by number of live births in the last 12 months, for Illubabor region, 1984 census

Age group	Number of Women	Live Births	ASFR per 100
15-19	26,484	2,760	104.2
20-24	27,484	6,355	231.2
25-29	33,608	8,355	248.6
30-34	30,205	6,958	230.4
35-39	29,012	6,524	224.9
40-44	22,757	3,722	163.6
45-49	16,982	2,139	126.0
Total	186,532	36,813	1,328.9

Based on the information and the tables given above,

Sex ratio (SR) = (Total male population/Total female population)*100

$$= (484637/491021)*100 = 987$$

Crude birth rate (CBR) = (36813/491021)*1000 = 74.97

Dependency Ratio (DR) = CDR+ADR

$$\begin{aligned} &= \left(\frac{P_{0-14}}{P_{15-64}} * 100 \right) + \left(\frac{P_{65+}}{P_{15-64}} * 100 \right) \\ &= ((18.4+18.2+10.4)/48.5)*100 + (4.5/48.5)*100 \\ &= (47/48.5)*100 + (4.5/48.5)*100 \\ &= 96.91+9.28 \\ &= 106.19 \end{aligned}$$

Crude death rate (CDR) = (Total number of deaths/Mid-year population)

$$\begin{aligned} &= (10462/975658)*1000 \\ &= 10.72 \end{aligned}$$

Calculate and **interpret** the following Demographic indices.

Infant mortality rate, maternal mortality rate, General fertility rate, age specific fertility rate, total fertility rate and gross reproduction rate child-woman ratio, rate of natural increase and doubling time.

7.5.3 Health Services Statistics

Health Service Statistics are very useful to improve the health situation of the population of a given country. For example, the following questions could not be answered correctly unless the health statistics of a given area is consolidated and given due emphasis.

1. What is the leading cause of death in the area? Is it malaria, tuberculosis, etc?
2. At what age is the mortality highest, and from what disease?

3. Are certain diseases affecting specified groups of the population more than others? (this might apply, for example, to women or children, or to individuals following a particular occupation)
4. In comparison with similar areas, is this area healthier or not?
5. Are the health institutions in the area able to cope with the disease problem?
6. Is there any season at which various diseases have a tendency to break out? If so, can these be distinguished?
7. What are the factors involved in the incidence of certain diseases, like malaria, tuberculosis, etc.?

Uses of Health Statistics

The functions/uses of health statistics are enormous. A short list is given below:

- 1) Describe the level of community health
- 2) Diagnose community ills
- 3) Discover solutions to health problems and find clues for administrative action
- 4) Determine priorities for health programmes
- 5) Develop procedures, definitions, techniques such as recording systems, sampling schemes, etc.
- 6) Promote health legislation
- 7) Create administrative standards of health activities
- 8) Determine the met and unmet health needs
- 9) Disseminate information on the health situation and health programmes
- 10) Determine success or failure of specific health programmes or undertake overall evaluation of public health work
- 11) Demand public support for health work

Major limitations of morbidity and mortality data from health institutions in Ethiopia include the following:

- 1) **Lack of completeness:** Health services at present (in 2000) cover only 47% of the population
- 2) **Lack of representativeness:** Illnesses and deaths recorded by health institutions do not constitute a representative sample of all illnesses & deaths occurring in the community.
- 3) **Lack of denominator:** The underlying population served by a health institution is difficult to define
- 4) **Lack of uniformity in quality:** No laboratory facilities in health stations. Such facilities are available in hospitals.
- 5) **Lack of compliance with reporting:** Reports may be incomplete, not sent on time or not sent at all.

Health service utilization rates (Hospital statistics) - Indices relating to the hospital

1. **Admission rate (AR):** The number of (hospital) admissions per 1000 of the population per year

$$AR = \frac{\text{Number of admission in the year}}{\text{Total Population of the catchment area}} \times 1000$$

“Admission” is the acceptance of an in-patient by a hospital.

Discharges and deaths: The annual number of discharges includes the number of patients who have left the hospital (cured, improved, etc.), the number who have transferred to another health institution, and the number who have died.

2. **Average length of stay (ALS):** This index indicates the average period in hospital (in days) per patient admitted. Ideally, this figure should be calculated as:

$$ALS = \frac{\text{The Annual Number of Hospitalized Patient Days}}{\text{Number of Discharges} \wedge \text{Deaths}}$$

That is, cumulative number of bed-days of all discharged patients (including those dying in hospital) during one year divided by the number of discharged and dead patients in the same year.

3. Bed-occupancy rate (BOR): This figure expresses the average percentage occupancy of hospital beds.

$$\text{BOR} = \frac{\text{The Annual Number of Hospitalized Patient Days}}{\text{Total number of beds}} \times \frac{1}{365}$$

4. Turnover interval (TI): the turnover interval expresses the average period, in days, that a bed remains empty, in other words, the average time elapsing between the discharge of one patient and the admission of the next.

$$\text{TI} = \frac{(365 \times \text{Number of Beds}) - \text{Number of Hospitalized Patient Days}}{\text{Number of Discharges} \wedge \text{Deaths}}$$

This figure is obtained by subtracting the actual number of hospitalization days from the potential number of hospitalization days in a year and dividing the result by the number of discharges (and deaths) in the same year. The turnover interval is zero when the bed-occupancy rate is 100%.

A very short or negative turnover interval points to a shortage of beds, whereas a long interval may indicate an excess of beds or a defective admission mechanism.

5. Hospital Death Rate (HDR)

$$\text{HDR} = \frac{\text{Total Number of Hospital Deaths} \in \text{a Given Period}}{\text{Number of Discharges} \in \text{the given period}} \times 1000$$

Exercise

1. A survey was conducted in town “X” in September 2003. It revealed a total population of 389,000 (196,500 males and 192,500 females). Total number of children born alive during the previous year was 16,400. The survey has also recorded a total of 5835 deaths (3200 males and 2635 females) during the same year. The survey has also identified the following deaths among the population.

Children < 1 month = 370

Children 1 month - 11 months = 1100

Children 1 year – 4 years = 1865

Children 5 – 4 years = 800

Persons 15 – 49 years = 700

Persons 50 years and above = 1000

Death of mothers during pregnancy and child birth = 130

Based on the above data provided, calculate the following measures of fertility and mortality. If there no data available to calculate any one of the following measures of fertility or mortality your answer has to be “Data Not Available”.

Crude Birth Rate

Crude Death Rate

General Fertility Rate

Infant Mortality Rate

Neonatal Morality Rate

Post Neonatal Rate

Sex Ratio at Birth

Under five morality Rate

Sex Specific Mortality Rate for Females

Sex specific Mortality Rate for Males

Maternal Mortality Rate

Total Fertility Rate

Gross Reproductive Rate